Jianjin Xu

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Education

Carnegie Mellon University — Pittsburgh, PA
Ph.D. in Robotics, Advisor: Prof. Fernando De la Torre Frade — 8/2023 –

Columbia University — New York, NY
M.S. in Computer Science, Advisor: Prof. Changxi Zheng — 8/2019 – 5/2021

Tsinghua University — Beijing, CN
B.Eng. in Computer Science, Advisor: Xiaolin Hu — 8/2015 – 7/2019

Research Interests

3D Human Avatar Generation, Video Diffusion Model for Human, Generative Model, 3D Computer Vision

Ongoing Projects

Efficient Human Video Model — 2026.9 –

- Studied pruning and autoregressive distillation of LTX 2.5 and Wan Animate 2 for human-specific domain.

Human Behavior Metric — 2026.9 –

- Developed evaluation metrics using Vision-Language Models to assess behavioral similarity and motion consistency in generated videos, calibrated against human user studies.

Feedforward Multiview Animatable Head Avatar Model — 2026.3 –

- Existing methods for building 3D head avatars heavily rely on FLAME estimation, which is limited in side views, extreme expressions and efficiency. On the other hand, diffusion-based methods are much slower than 3D methods in inference and requires much larger networks.
- We build a new network that seek to strike a balance between the 3D consistency and inference efficency. We propose to unify multi-view synthesis and monocular animation with a feedforward network without diffusion.

Publications and Manuscripts

1. **Jianjin Xu**, Soham Dasgupta, Fernando De la Torre. ARG-Avatar: Animation-Rendering Guidance for Lightweight Feedforward Avatar Reconstruction. (ArXiv, 2026)
2. Li, Hengjia and **Jianjin Xu** and Cheng, Keli and Wang, Lei and Bi, Ning and Wu, Boxi and De la Torre, Fernando and Cai, Deng. Multi-View Consistent Human Image Customization via In-Context Learning. (ArXiv, 2025)
3. Lin, Yukang and Fung, Hokit and **Jianjin Xu** and Ren, Zeping and Lau, Adela SM and Yin, Guosheng and Li, Xiu. Mvportrait: Text-guided motion and emotion control for multi-view vivid portrait animation. (CVPR 2025)
4. Lin, Yuchen and Lin, Chenguo and **Jianjin Xu** and MU, Yadong. OmniPhysGS: 3D Constitutive Gaussians for General Physics-Based Dynamics Generation. (ICLR 2025)
5. Chen, Anthony and **Jianjin Xu** and Zheng, Wenzhao and Dai, Gaole and Wang, Yida and Zhang, Renrui and Wang, Haofan and Zhang, Shanghang. Training-free regional prompting for diffusion transformers. (ArXiv, 2024)
6. Kwon, Youngjoong and Fang, Baole and Lu, Yixing and Dong, Haoye and Zhang, Cheng and Carrasco, Francisco Vicente and Mosella-Montoro, Albert and **Jianjin Xu** and Takagi, Shingo and Kim, Daeil and others. Generalizable human gaussians for sparse view synthesis. (ECCV 2024)
7. **Jianjin Xu**, Saman Motamed, Praneetha Vaddamanu, Chen Henry Wu, Christian Haene, Jean-Charles Bazin, Fernando de la Torre. Personalized Face Inpainting with Diffusion Models by Parallel Visual Attention. (WACV 2024)
8. Saman Motamed, **Jianjin Xu**, Chen Henry Wu, Fernando De la Torre. PATMAT: Person Aware Tuning of Mask-Aware Transformer for Face Inpainting. (ICCV 2023)
9. **Jianjin Xu**, Zhaoxiang Zhang, Xiaolin Hu. Extracting Semantic Knowledge from GANs with Unsupervised Learning. (TPAMI, 2023)
10. **Jianjin Xu**, Changxi Zheng. Linear Semantics in Generative Adversarial Networks. (CVPR 2021)
11. **Jianjin Xu**, Zheyang Xiong, Xiaolin Hu. Frame Difference-Based Temporal Loss for Video Stylization. (ArXiv)

Professional Experience

Unrestricted Vehicle Re-Identification System with Deep Metric Learning 6/2018 – 10/2018

Internship at MSRA, supervised by Lead Researcher Xun Guo

- Developed a re-identification system that inputs raw videos of monitors and identifies re-appeared vehicles. The system first detects vehicles by faster RCNN, then conducts tracking and matching by learned deep metrics.
- Trained the deep metric model on VeRi dataset and validated it on VID dataset and collected traffic videos.

Awards

3rd Prize in 36th the Challenge Cup Competition, Tsinghua University 4/2018

2nd Prize in Mathematical Contest in Modeling, 2017 2/2017

Teaching Experience

TA @ Carnegie Mellon University, 16-822: Geometry-based Methods in Vision, Fall 2024 & 2025

TA @ Columbia University, COMS-W4995 Special Topics In Computer Science, I: Causal Inference, 2020

Miscellaneous Experience

Chairman of Tsinghua Future Internet and Computation Club 6/2018-6/2019

Skills

pytorch / tensorflow / python / C++ / javascript / CUDA